MIL-T-53023A(ME)
3 April 1985
SUPERSEDING
MIL-T-53023(ME)
21 March 1983

MILITARY SPECIFICATION

TACTICAL WATER DISTRIBUTION EQUIPMENT SYSTEM (TWDS-SETS), 10 MILE SEGMENT

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tactical water distribution set for use to distribute large quantity of potable water over great distances.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-A-8421 - Air Transportability Requirements, General Specifications for.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-23377 - Primer Coating, Epoxy Polymide, Chemical and Solvent Resistant.
MIL-S-81733 - Sealing and Coating Compound, Corrosive Inhibitive.

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1474 - Noise Limits for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS

Title 21 - Federal Food, Drug and Cosmetic Food Additives

(The Code of Federal Regulations is available from the Superintendent of Documents, US Printing Office, Washington, DC 20402.)

DRAWINGS

ME

TA13225E9090 - Tactical Water Distribution Equipment System (TWDS-SET) 10 Mile Segment.

(Copies of specifications, standards, drawings, and pubications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL SANITATION FOUNDATION STANDARD

Standard No. 14 - Plastic Piping System Components and Related Materials.

(Application for copies should be addressed to National Sanitation Foundation, NSF Building, Ann Arbor, MI 48015.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

AMERICAN WATER WORKS ASSOCIATION

(Application for copies should be addressed to the American Water Works Association, 2 Park Avenue, New York, NY 10016.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The Tactical Water Distribution Equipment System (TWDS-SET) 10 Mile Segment, hereinafter referred to as TWDS, shall be as specified in components list (see appendix) and TA13225E9090. The major components of TWDS include ten miles of hoseline, six pumps, two storage tanks, two distribution points, and necessary hose connecting hardware to permit assembly of the set components.

3.1.1 Drawings. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrrect fits, the contractor shall provide tolerances within those prescribed on the drawing to insure correct fit, assembly, and operation of the TWDS. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from

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a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of TWDS from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish TWDS conforming to this specification.

3.3 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 3.3.4). Materials in contact with potable water shall be certified in accordance with the current applicable American Water Works Association Standards, National Sanitation Foundation Standard No. 14 for plastic piping system components and related materials, or Code of Federal Regulations, Title 21, Parts 177 and 182 for rubber compounds in contact with potable water.

3.3.1 Material deterioration and control. The TWDS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889 shall be electrically insulated from one another to minimize galvanic corrosion. Insulation shall be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 against any galvanic corrosion shall also be obtained by exculsion of the electrolyte if feasible.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the TWDS may be newly fabricated from recovered materials to the maximum extent practicable, provided the TWDS produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the TWDS.

3.4 Components. The sizes, types and quantities of components shall be as specified in components list (see appendix) and TA13225E9090.

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3.5 Treatment and painting. The portions of the TWDS normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G as applicable, with a finish coat color sand.

3.6 Metalic component finish. Aluminum parts shall have a dyed, anodized coating conforming to MIL-A-8625, type II, class 2, color to approximate sand matte.

3.7 Government-furnished property. When specified (see 6.2) the following property in the quantities indicated will be furnished by the Government (see 6.4):

	Description	Identification	Quanity
a.	Bag, drinking water storage	4610-00-268-9890	4 ea.
b.	Comparator, color w/case	6630-01-044-0334	2 ea.
c.	Hypochlorination unit, 2-100 gpm	4610-00-880-0154 or 4610-00-937-6862	2 ea.
d.	Pump, 125 gpm	4320-00-542-3347	2 ea.
e.	Tank, Fabric, Collapsible, 20,000 gal	5430-01-106-9678	2 ea.

3.8 Workmanship. All parts and components of the TWDS shall be clean and free from extraneous material. Any necessary rework shall be limited to procedures which will not affect the strength or functioning of parts or components.

3.9 Safety and noise.

3.9.1 Safety. Nonfunctional sharp edges, projecting points and excessive lengths of fastening devices shall be avoided.

3.9.2 Noise limits. The noise produced by the TWDS components shall conform with MIL-STD-1474 requirements when tested in accordance with 4.5.2.2. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirments specified herein are as follows:

a. First article TWDS inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article TWDS inspection.

4.3.1 Sampling. The first article items of each component listed in the components list shall be provided for examination and test.

4.3.2 Examination. The first article components provided in accordance with 4.3.1 shall be examined in accordance with the applicable specification requirements for first article examination and as specified in 4.5.1. Components not covered by a specification shall be examined as specified in 4.5.1 and for compliance with applicable MS part number or drawing refered to in the component list. Presence of one or more defects in a component shall be cause for rejection of that component.

4.3.3 Tests. The first article components provided in accordance with 4.3.1 covered by a specification shall be tested as specified in 4.5.2.1 and 4.5.2.2. Rejection criteria shall be in accordance with the requirements of the applicable component specification referenced in the components list.

4.4 Quality conformance inspection.

4.4.1 Sampling. Samples shall be taken in accordance with the applicable specification for each component listed in the components part list. For components not covered by a specification, samples shall be selected in accordance with MIL-STD-105, AQL 2.5, for examination and testing.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined in accordance with the applicable specification requirement and as specified in 4.5.1. Components not covered by a specification shall be examined as specified in 4.5.1.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. Rejection criteria shall be in accordance with the requirements of the applicable specification referenced in the components list. Selected samples not covered by a specification shall meet the requirements of the applicable MS part number or drawing cited in the components list.

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4.5 Inspection procedure.

4.5.1 Examination. The TWDS shall be examined as specified herein for the following defects:

101. Material not as specified or not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
102. Contractor does not have documentation available for identification of material, material finishes or treatments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the TWDS.
105. Components size, type, and quantity not as specified in components list (see appendix). Each incorrect component size, type or quantity shall constitute one defect.
106. Treatment and painting not as specified.
107. Metalic component finish not as specified.
108. Workmanship not as specified.
109. Safety not as specified.
110. Certification that material in contact with potable water is in accordance with current applicable standards governing the handling of potable water not furnished (see 3.3).

4.5.2 Tests.

4.5.2.1 Component test. Unless otherwise specified (see 6.2), each component shall be tested and accepted in accordance with the requirments of the applicable component specification referenced in the components list.

4.5.2.2 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load; MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position; additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination Examine the first article pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test.

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4.6.1.2.1 Group 1, 2, 3, and 4 (see 5.3.1.1). The first article pack of crated components in group 1, 2, 3, and 4 (see 5.3.1.1) for level A shall be subjected to one of the tests specified herein. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing or any visual damage to the components shall be cause for rejection.

a. Rail-impact test. Crated components shall be subjected to the rail-impact test in accordance with MIL-STD-1186, appendix A. Crates not exceeding 9.5 feet in length shall be subjected to tests on each end and each side, a total of 8 impacts.
b. Pendulum impact test. Crated components shall be subjected to the pendulum impact test in accordance with FED-STD-101, method 5012; except the vertical height of drop shall be 18 inches.

4.6.1.2.2 Group 6, 7, and 8 (see 5.3.1.1). The first article pack of crated components in group 6,7, and 8 (see 5.3.1.1) for level A shall be subjected to the test as required by the applicable component specifications.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection a completed pack of each group of components (see 5.3.1.1 and 5.3.1.2) prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110. Components procured by a specification not preserved and packed as required by the applicable documents, for level A.
111. Components not procured by a specification not preserved as specified for level A.
112. Preservative that will adversely affect potable water applied to surfaces that will contact potable water for level A or commercial.
113. Consolidation not as specified for level A.
114. Boxes not as specified for level A.
115. Components not formed in groups as specified for level A.
116. Crates not constructed in accordance with referenced documents as specified for level A.
117. Blocking, bracing, and anchoring not in accordance with the referenced document for level A.
118. Crate size and weight not compatible with the air transportability requirements specified for level A.
119. Marking missing, illegible, incorrect or incomplete for level A or commercial.

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5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approved by the Government (see 6.5). The first article pack may be accomplished utilizing either the first article model or production item. If the first article model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model and the production items.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Components of the sets procured by a federal or military specification shall be preserved in accordance with the level A requirements of the applicable document. All other components of the set shall be preserved as specified for components of similar design and construction to prevent deterioration or damage during shipment, multiple handling and indeterminate storage, using the criteria established in MIL-P-116 as guidance. Specifically no preservation that will adversely affect potable water shall be applied to any surface that will contact potable water.

5.2.1.1 Consolidation. Components of the set shall be consolidated together in groups, in a manner best suited to accomplish the packing specified in 5.3. To aid in accomplishing that packing, small components shall be individually placed in, or a number of small components placed in, a close fitting box or boxes conforming to PPP-B-636, class weather resistant. To the greatest extent practical, components contained together in the same box shall be of like description, an assembly, set, etc. The contents shall be immobilized within the box to prevent movements or damage. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification. The boxes shall then be consolidated together in a minimum number of boxes conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. The boxes shall be of a size and configuration best suited to accomplish the packing specified in 5.3. Large components requiring the protection afforded by a wood box shall be either individually or in groups, sets, assemblies, etc., placed directly in a wood box and the contents immobilized as specified herein. Strapping of boxes shall not be required.

5.2.2 Commercial. Components of the set procured by a federal or military specification shall be preserved in accordance with the commercial requirements as contained in the applicable specification. All other components of the set shall be preserved in accordance with ASTM D 3951. Specifically no preservative that will adversely affect potable water shall be applied to any surface that will contact potable water.

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5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Grouping. Components of the set, preserved as specified in 5.2, shall be formed into eight separate groups as follows: Group 1, shall consist of all components comprising the pumping station except for the 600 GPM pumping assembly. Group 2, shall consist of all components comprising the storage assembly except for the 20,000 gallon collapsible tank. Group 3, shall consist of all components comprising the distribution point except for the 125 GPM pump. Group 4, shall consist of all components comprising the 10 mile segment except for the 6-inch, 500 foot length hose assembly. Group 5, shall consist of the 600 GPM pumping assembly. Group 6, shall consist of the 20,000 gallon collapsible tank. Group 7, shall consist of the 125 GPM pump. Group 8 shall consist of the 6-inch, 500 foot length hose assembly. Components in one group shall not be intermixed with components from another group.

5.3.1.2 Packing. Components of the set, segregated into groups as specified in 5.3.1.1, shall be packed as follows: Groups 1, 2, 3, and 4 shall be packed in a minimum number of crates conforming to MIL-C-104, type II, class 1 or 2, style A. The size and weight limitation of each crate shall be compatible with the limitations for C-130 aircraft. (See MIL-A-8421 for air transportability requirements.) The contents of the crate shall be blocked, braced and anchored in accordance with MIL-STD-1186. Groups 5, 6, 7, and 8 shall be packed in accordance with the level A packing requirements of the specification applicable to the item in the group.

5.3.2 Commercial. Components of the set, procured by a federal or military specification shall be packed in accordance with either the level C or commercial requirements as contained in the applicable specification. All other components of the set shall be packed in accordance with ASTM D 3951.

5.4 Marking. Marking for military levels of protection (level A) shall be in accordance with MIL-STD-129. Marking for commercial shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The tactical water distribution set (a set is a 10 mile segment of a 70-mile system) is used in those areas of the world where sources of water are extremely limited and situated at considerable distances from areas of consumption.

6.2 Ordering data.

a. Title, number and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of the first article TWDS and number of TWDS required (see 3.2).

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d. When components shall not be tested (see 4.5.2.1).
e. When Government-furnished property is required (see 3.7).
f. Time frame required for submission of the first article pack (see 5.1).
g. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the items should be an initial production TWDS. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first article.

6.4 Government-furnished property. The contracting officer shall arrange to furnish the property specified in 3.7, when applicable (see 6.2).

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preseved, pack, and mark the components in accordance with this specification.

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APPENDIX

COMPONENTS LIST

10. SCOPE

10.1 Scope. This appendix gives details of the components that comprises one Tactical Water Distribution Equipment System (TWDS-SET) 10 Mile Segment. Drawing 13225E9090 shows one possible layout the listed components can be assembled. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications and standards.

SPECIFICATIONS

FEDERAL

GGG-W-641	- Wrench, Socket (and Sockets, Handles, and Attachments for Socket Wrenches, Hand).

MILITARY

MIL-B-273	- Bag, Drinking Water Storage, Dispensing, Cloth, Cotton Duck, Porous.
MIL-C-538	- Comparator, Color.
MIL-C-10387	- Coupling, Clamp, Pipe, with Bolts and Synthetic Rubber Gaskets for Grooved End Pipe.
MIL-P-10388	- Pipe Fitting, One or More Ends Grooved.
MIL-H-12732	- Hypochlorination Unit, Water Purification, Frame Mounted, Automatically Contolled 2 to 100 GPM Flow.
MIL-C-27487	- Coupling Halves, Quick-Disconnect, Cam-Locking Type.
MIL-P-52109	- Pump, Centrifugal, 2-Inch, Gasoline-or Diesel-Engine-Driven.
MIL-S-53012	- Suspension Kit, 6-Inch Hoseline.
MIL-R-53013	- Repair Kit, 6-Inch Hoseline.
MIL-S-53014	- Sling Assembly: Flaking Box, Lighting, 6-Inch Hoseline.
MIL-D-53015	- Displacement and Evacuation Kit, 6-Inch Hoseline.

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APPENDIX

MIL-P-53016	- Packing Kit, 6-Inch Hoseline.
MIL-G-53017	- Guard, Roadway Crossing, 6-Inch Hoseline.
MIL-H-53027	- Hose Assembly, Rubber: Lightweight, Collapsible, 6-Inch; for Drinking (Potable) Water.
MIL-T-53029	- Tanks, Fabric, Collapsible, Pillow Type: 10,000, 20,000, and 50,000 Gallon, Potable Water.
MIL-R-82184	- Regulator Assembly, Skid Mounted, 6-Inch.

STANDARDS

MILITARY

MS27020	- Coupling Half, Quick Disconnect, Cam and Locking Type, Male, Internal Pipe Thread, Type I.
MS27022	- Coupling Half, Quick Disconnect, Cam and Locking Type, Male, External Pipe Thread, Type III.
MS27023	- Coupling Half, Quick Disconnect, Cam and Locking Type, Male, Flanged, Type IV.
MS27024	- Coupling Half, Quick Disconnect, Cam and Locking Type, Female, Internal Pipe Thread, Type V.
MS27026	- Coupling Half, Quick Disconnect, Cam and Locking Type, Female, External Pipe Thread, Type VII.
MS27027	- Coupling Half, Quick Disconnect, Cam and Locking Type, Female, Flanges, Type VIII.
MS27029	- Coupling Half, quick Disconnect, Cam and Locking Type, Plug Dust, Type X.
MS49000	- Reducer, Male by Female and Female by Male, Quick Disconnect, Cam and Locking, Type XI.
MS49002	- Reducer, External Pipe Thread by Quick Disconnect Cam-Locking Type, Female, Type XIII.
MS90714	- Nail, Composition Roofing.

20.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J104 - Square Hex Nuts.
SAE J105 - Hex Bolts.
SAE J488 - Plain Washers.
SAE J489 - Lock Washers.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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National Stock Number	Description	U/I	Qty
4320-01-175-3933	Tactical Water Distribution Equipment System (TWDS-SET) 10 Mile Segment; 97403---13225E9090 Consisting of the following components:		
	Coupling, Clamp, Pipe: 6 in. nom. size (2 coupling halves, 1 gasket for potable water, 2 bolts and nuts; (81349) MIL-C-10387, C10387-1-D-11	ea.	15
	Coupling Half, Quick Disconnect: Cam-locking type, female, IPT, type V, alum, 6 in. nom. size; (96906) MS27024-19	ea.	1
	Displacement and Evacuation Kit, Hoseline: 6 in. nom. hose size; (81349) MIL-D-53015; (97403) dwg 13226E1577	ea.	2
	Guard Road Crossing: (81349) MIL-G-53017; (97403) dwg 13226E1576	ea.	24
	Hose Assembly, Rubber: Lightweight collapsible, 6 in., 500 ft. long, Victualic type end fittings, 1 coupling and flaking box; (81349) MIL-H-53027	ea.	128
	Hose Assembly, Rubber, Smooth bore: Potable water suction, 6 in. nom. size 10 ft. long, with quick-disconnect, cam-locking fittings; (81349) MIL-C-27487, Type II and Type VI, Class 1; (97403) dwg 13225E9135-6	ea.	12
	Lubricant, Gasket: for coupling assembly, one-quart can, (97403) dwg 13225E9192	ea.	2

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National Stock Number	Description	U/I	Qty
4320-01-175-3933	Tactical Water Distribution Equipment System (TWDS-SET) 10 Mile Segment; (Cont'd)		
	Nails: Composition Roofing, .135 in. dia., 1 1/2-inch long; (96906) MS90714-8	ea.	240
	Packing Kit, Hoseline: 6 in. nom. hose size; (81349) MIL-P-53016; (97403) dwg 13226E1571	ea.	1
	Pipe Fitting, Cap: 6 in. nom. size, alum.; (81349) MIL-P-10388, M10388-A-05-AK-1-C	ea.	1
	Pipe Fitting, Coupling: External threaded one end, alum., 6 in.; (81349) MIL-P-10388, M10388-A-07-AK-1-C-6A	ea.	1
	Relief Valve Assembly: (97403) Dwg 13225E9196	ea.	1
	Repair Kit, Hoseline: 6 in. nom. hose size; (81349) MIL-R-53013; (97403) dwg 13226E1581	ea.	1
	Skid: For pressure valve assembly, to fit and assemble with Cla-Val Co. part No. 6"-906-01BV Victualic ends; iaw paragraph 3.3.3 of MIL-R-82184	ea.	1
	Sling Assembly: Four legs with spreader bar; (81349) MIL-S-53014; (97403) dwg 13226E1582	ea.	1
	Suspension Kit, Hoseline: 6 in. nom. hose size; (81349) MIL-S-53012	ea.	5
	Swivel Joint: 6 in., Victalic type end fittings, with 1 coupling; (97403) dwg 13225E9199	ea.	64
	TWDS (RDF) Pumping Station; (97403) dwg 13225E9088; consisting of:	ea.	6

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National Stock Number	Description	U/I	Qty
4320-01-175-3933	Tactical Water Distribution Equipment System (TWDS-SET) 10 Mile Segment; (Cont'd)		
	8 ea. Coupling, Clamp, Pipe: 6 in. nom. size (2 coupling halves, 1 gasket for potable water 2 bolts and nuts; (81349) MIL-C-10387, C10387-1-D-11		
	1 ea. Hose Assembly, Rubber: Lightweight, collapsible, 6 in., 20 ft. long, Victualic type end fittings, 1 coupling; (81349) MIL-H-53027		
	1 ea. Hose Assembly, Rubber: Lightweight, collapsible 6-inch, 50 ft. long, Victualic type end fittings, 1 coupling; (81349) MIL-H-53027		
	1 ea. Hose Assembly, Rubber: Lightweight, collapsible, 6-inch, 10 ft. long, Victualic type end fittings 1 coupling; (81349) MIL-H-53027		
	2 ea. Lateral, 45 degrees: All ends grooved, 6-inch nom. size, alum; (81349) MIL-P-10388, M10388-A-18-AK-1-C		
	1 ea. Pumping Assembly: 600 gpm @ 350 tdh, DED, wheel mounted, 6 x 6 Victualic connections, intervehicular connections for 24-volt electric system; (97403) DWG 13225E9180		
	1 ea. Relief Valve Assembly: (97403) Dwg 13225E9196		

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National Stock Number	Description	U/I	Qty
4320-01-175-3933	Tactical Water Distribution Equipment System (TWDS-SET) 10 Mile Segment; (Cont'd)		
	3 ea. Valve, Butterfly: 6 in. standard trim, EPDM, with gear actuator, Victualic ends; (79154) series 700		
	1 ea. Valve, Check: EPDM, 6 in., Victualic ends; (79154) Style 710.		
	TWDS (RDF) Distribution Point: (97403) dwg 13225E9091; consisting of:	ea.	2
4610-00-268-9890	2 ea. Bag, Drinking Water Storage: Dispensing cloth, cotton duck, porous; (81349) MIL-B-273, (97403) dwg 13200E6480		
6630-01-044-0334	1 ea. Comparator Color: with case, (81349) MIL-C-538, (97403) dwg 13200E7400 and dwg 13216E9016		
4730-00-088-9285	4 ea. Coupling Half, Quick Disconnect: Cam-locking, female, external pipe thread, type VII, alum, 2 in. nom. size; (96906) MS27026-11		
4730-00-938-7997	4 ea. Coupling Half, Quick Disconnect: Cam-locking, male, external pipe tread, type III, alum., 2 in. nom. size; (96906) MS27022-11		

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4730-00-915-5127

2 ea. Coupling half, Quick Disconnect: Cam-locking type, plug, dust, type X, 2 in., (96906) MS27029-11

1 ea. Hose Assembly, Rubber, Smooth Bore: Potable water discharge, 2 in. nom. size, 10 ft. long, with quick-disconnect, cam-locking fittings (81349) MIL-C-27487, Type II and VI, Class 1 (1 male, 1 female); (97403) dwg 13225E9136-1

7 ea. Hose Assembly, Rubber, Smooth Bore: Potable water discharge, 2 in. nom. size, 20 ft. long, with quick-disconnect cam-locking fittings (81349) MIL-C-27487, Type II and VI, Class 1 (1 male, 1 female); (97403) dwg 13225E9136-2

7 ea. Hose Assembly, Rubber, Smooth Bore: Potable water discharge, 2 in. nom. size, 20 ft. long, with quick-disconnect cam-locking fittings (81349) MIL-C-27487, Type II and VI, Class 1 (1 male, 1 female); (97403) dwg 13225E9136-2

1 ea. Hose Assembly, Rubber, Smooth Bore: Potable water suction, 2 in. nom. size, 20 ft. long, with quick-disconnect cam-locking fittings (81349) MIL-C-27487, Type II and VI, Class 1 (1 male, 1 female); (97403) dwg 13225E9135-2

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National Stock Number	Description	U/I	Qty
4610-00-880-0154 4610-00-937-6862	1 ea. Hypochlorination Unit: Water purification, frame mounted, water driven, 100 gpm; (81349) MIL-H-12732; (09680) 1955-1; (79172) A-506131		
	2 ea. Nozzle: Alum. body with spout, internal pipe thread, 1-1/2 in. nom. size; (97403) dwg 13225E9096		
4320-00-542-3347	1 ea. Pump: Centrifugal, 2 in., GED 125 gpm @ 50 ft. head; (97403) dwg 13200E8800; (81349) MIL-P-52109, Class 1		
4730-00-951-3295	2 ea. Reducer: Female by male, quick-disconnect, cam-locking, type XI, alum., 2 x 1-1/2 in. nom. size; (96906) MS49000-5		
4730-01-064-0560	1 ea. Reducer: Female by male, quick-disconnect, cam-locking type XI, alum., 4 x 2 in. nom. size; (96906) MS49000-17		
	2 ea. Reducer: External pipe thread by quick-disconnect, cam-locking type, female, type XIII, alum., 2 in. nom. size by 1-1/2 - 11-1/2 NPT; (96906) MS49002-9		
	1 ea Reducer: Female by male, quick disconnect, cam-locking, type XI, alum, 2 x 4-inch nom size; (96906) MS49000-19		
	1 ea Reducer: Female by male, quick-disconnect, cam-locking, type XI, alum, 2 x 3-inch nom size; (96906) MS49000-11		

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National Stock Number	Description	U/I	Qty
	2 ea. Stand Assembly, Nozzle; (97403) dwg 13225E9140		
	2 ea. Valve: Elbow coupler, quick-disconnect, cam-locking type, female; 2 in. nom. size; for potable water; (97403) dwg 13219E0491		
4820-00-810-4250	2 ea. Valve, Gate: Bronze, threaded, 2-11-1/2 NPT int; (82666) 2 inch PN B120		
4730-01-068-5070	3 ea. Y-connection, Quick-Disconnect: Cam-locking, male-male-female, alum., 2 in. nom. size; (97403) Dwg 13219E0477		
	TWDS (RDF) Storage Assembly: (97403) dwg 13225E9089; consisting of:	ea.	2
	1 ea. Coupling, Clamp, Pipe: 6 in. nom. size (2 coupling halves, 1 gasket for potable water, 2 bolts and nuts; (81349) MIL-C-10387, C10387-1-D-11		
	1 ea. Coupling, Clamp, Pipe: 4 in. nom. size (2 coupling halves, 1 gasket for potable water, 2 bolts and nuts; (81349) MIL-C-10387, C10387-1-D-09		
	1 ea. Coupling Half, Quick-Disconnect: Cam-locking, male, internal pipe thread, type I, alum., 4 in. nom. size; (96906) MS27020-17		
	1 ea. Coupling Half, Quick Disconnect: Cam-locking type, female, flanged, type VIII, alum, 4-in. nom. size; (96906) MS27027-17		

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National Stock Number	Description	U/I	Qty
	1 ea. Coupling Half, Quick-Disconnect: Cam-locking type, male, flanged, type IV, alum, 4-in. nom. size; (96906) MS27023-17		
	2 ea. Gasket: (97403) Dwg 13220E1069		
	4 ea. Hose Assembly, Rubber, Smooth Bore: Potable water discharge, 4 in. nom. size, 10 ft. long, with quick-disconnect, cam-locking fittings; (81349) MIL-C-27487, Type II and Type VI, Class 1 (1 male, 1 female); (97403) dwg 13225E9136-3		
	16 ea. Nut, Plain: Hexagon-carbon steel, cadmium plated, .375-16UNC-2B; (81343) SAE J104		
	1 ea. Pipe Fitting, Coupling: External threaded one end, alum., 4 in. nom. size; (81349) MIL-P-10388, M10388-A-07-AI-1-C-6A		
	1 ea. Pipe Fitting, Tee: Reducing grooved all ends, 6x6x4 in. nom. size, alum.; (81349) MIL-P-10388, M10388-A-30-EB-1-A-6C		
5430-01-106-9678	1 ea. Tank: Fabric, collapsible, pillow type, 20,000 gal. capacity (for potable water); (81349) MIL-T-53029		
	16 ea. Screw, Cap: Hexagon head (finished hexagon bolt), steel, grade 5, cadmium plated, .375-16UNC-2A x 3.250 long; (81343) SAE J105		

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National Stock Number	Description	U/I	Qty
	1 ea. Valve, Gate: flanged, 4 in. nom. size; (97403) dwg 13220E1071		
	32 ea. Washer: Flat-round, steel, cadmium plated, general purpose, .406 basic ID x .812 basic OD; (81343) SAE J488		
	16 ea. Washer: Lock-spring, helical, regular (medium) series, steel, cadmium plated, .375 nom. size; (81343) SAE J 489		
	Valve Assembly: Pressure reducing, potable water, 6 in. Victualic connections, assembled with the skid; (86184) PN 6"-90G-01BV Victualic ends	ea	1
	Wrench, Socket: Style A, 3/4-inch drive, 1.06 nom; (81348) GGG-W-641, type II, CL 2	ea	2
	Wrench, Socket: Style A, 3/4-inch drive, 1.25 nom; (81348) GGG-W-641, type II, CL 2	ea	4
	Wrench, Socket, Handle: 3/4-inch drive; (81348) GGG-W-641, type III, CL2	ea	4
	Y-Connection, Quick-disconnect; (97403) 13225E9190: Consisting of:	ea	1
	1 ea. Coupling Half, Quick-disconnect: Cam-locking type, male, NPT, type I, alum, 6-inch nom size; (96906) MS27020-19		

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1 ea. Coupling: EPT one end, alum, 6-inch nom size; (81349) MIL-P-10388, M10388-A-07-AK-1-C-6A

2 ea. Coupling Half, Quick-disconnect: Cam-locking type, female, NPT, type V, alum, 4-inch nom. size; (96906) MS27024-17

3 ea. Coupling, Clamp, Pipe: 6-inch nom size (2 coupling halves, 1 gasket for potable water, 2 bolts and nuts); (81349) MIL-C-10387, C10387-1-D-11

2 ea. Reducer: alum, threaded small end, 6 x 4-inch nom size; (81349) MIL-P-10388, M10388-A-28-BT-1-C-4A

1 ea. Y-Connection: Grooved all ends, alum, 6-inch nom. size; (97403) dwg 13225E9191

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-53023A(ME)	Tactical Water Distribution Equipment System (TWDS-SETS), 10 Mile Segment

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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